

```

~ sudo service rundeckd start
~ sudo service rundeckd status
rundeckd.service - LSB: rundeck job automation cor
Loaded: loaded (/etc/init.d/rundeckd; bad; vendor
Active: active (running) since Thu 2019-03-07 02:
Docs: man:systemd-sysv-generator(8)
Process: 7546 ExecStop=/etc/init.d/rundeckd stop
Process: 7635 ExecStart=/etc/init.d/rundeckd start
Main PID: 7649 (su)
Tasks: 0
Memory: 696.0K
CPU: 28ms

```

Figure 6: Rundeckd daemon status

regarding the status of the jobs being executed. The workflow of this is shown in Figure 1.

Installation of Rundeck

Figure 5 gives the flowchart for the installation procedure of Rundeck on Linux systems.

Now, let us go through the installation procedure in detail. A Rundeck instance can be installed on systems with a Linux OS in four easy steps as described below. Rundeck is also compatible with Windows and Mac but this article focuses only on the procedure for Linux systems.

1. Install Java, if not already installed, using the following command:

```
$ sudo apt install openjdk-8-jdk
```

2. Get the latest Rundeck package from docs.rundeck.com/download/deb/ and install it using the command below. *Dpkg* is a package manager for Debian based systems and it requires root access. The '-i' option is to install a given package.

```
$ sudo dpkg -i { .deb file name }
```

3. Rundeck is now successfully installed on the system. *Rundeckd* is the daemon process that manages the Rundeck server. Start the Rundeck daemon using the following command and verify that the service is running.

```
$ sudo service rundeckd start [6]
```

4. The Web UI of the Rundeck instance can be accessed from *localhost:4440/user/login*. The default port used by the Rundeck

Create a new Project

Project Name
Demo

Description
This is a demo project

Execution Mode
Job execution and schedule configuration at project level.

☐ Disable Execution

☐ Disable Schedule

User Interface
Additional configuration for the user interface for this project

Job Group Expansion Level 1
In the Jobs page, expand Job groups to this depth by default. [More >](#)

Figure 7: New project creation

Node Sources: Demo

Project Nodes
Additional configuration for the Nodes for this project

☒ Use Asynchronous Cache
Use asynchronous cache for all Resource Model Source results in this project

Cache Delay 30
Delay in seconds, at least 30. [More >](#)

You can add additional custom sources, and their results will be used with the ordering shown. Later sources will override earlier sources. (You can use `$(project.name)` inside configuration values to substitute the project name.)

Order	File	File Path	Generate	Include Server Node
1.	File Reads a file containing node definitions in a supported format	/var/rundeck/projects/Demo/etc/resources.xml	Yes	Yes

[Delete](#) [Edit](#)

[Add Source +](#)

[Cancel](#) [Save](#)

Figure 8: Node sources tab

Web interface is 4440. We can log in using 'admin' for the user name as well as the password. Once we log in as the admin, we can also create other users with different permissions and authorisations. Login page it displays already existing projects, as well as the options to configure these projects and to add a job in them. It also shows the number of jobs executed in the last one day.

Since Rundeck executes in Linux as a service, we can control the service by using the *start*, *stop*, *restart* and *status* commands, like with any other service.

```
$ sudo service rundeckd stop/ start/ restart/ status
```

Now that we have the Rundeck server running, let us go through the details of how to add a new project, create and configure jobs, how to add nodes, and how to automatically deploy a program or software onto the nodes.

Dispatching code to the nodes using Rundeck

1. Creating a new project

On the Rundeck Web UI, click the 'New Project' button that is shown in Figure 8. Enter the 'Name' and 'Description'